Gas exchange in leaves

9B Plant growth · Lesson 3 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can describe how stomata and guard cells control gas exchange in a leaf.
- ☐ I can explain how temperature and concentration difference affect diffusion, and why a plant loses more water on a hot day.

What happens today

1. Do Now: a leaf under the microscope
2. Learn: Stomata and guard cells · Diffusion · and more
3. Practical: Looking at stomata
4. Challenge, then the exit ticket

Student Book pages 25 · Workbook pages 18

What I learned today (one or two sentences) 今天我学到了：

Key Vocabulary

Term	中文	Definition
stomata	气孔	tiny holes in a leaf that let gases in and out (one is a stoma)
guard cell	保卫细胞	one of the pair of cells that opens and closes a stoma
diffusion	扩散	the overall movement of particles from where they are more concentrated to where they are less concentrated
concentration	浓度	how many particles of a substance there are in a certain space
gas exchange	气体交换	the swapping of gases between a leaf and the air
water vapour	水蒸气	water in the form of a gas
evaporation	蒸发	a liquid turning into a gas
wilt	枯萎	to droop because the cells have too little water
epidermis	表皮	the outer layer of cells of a leaf

Command words 指令词: Complete 完成（填写） · Draw 画出 · Estimate 估算 · Explain 解释 · Give 给出 · Give a reason 给出理由 · Label 标注 · Name 说出名称 · State 说明（写出） · Suggest 提出（建议） · Tick 打勾 · Use 使用（运用）

Quick Review

a leaf under the microscope

1 What is this hole in the leaf surface called?

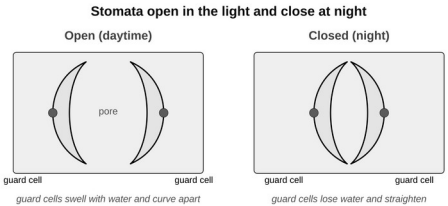
2 Name two gases that pass through it.

3 Predict when the hole is shut: day or night?

Core 1 of 5

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 The diagram shows a stoma in the lower epidermis of a leaf at two different times of day.



a) State the function of the guard cells.

Core 2 of 5

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 *continued*

b) Name two gases that are exchanged between the leaf and the air through stomata.

c) Which diagram shows a stoma at night? Give a reason for your answer.

Core 3 of 5

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

2 Complete the sentences. Use the word bank: concentration, fall, photosynthesis, rapidly, respiration, rise, slowly.

In bright light, _____ takes up carbon dioxide from the air inside the leaf.

This causes the _____ of carbon dioxide inside the leaf to _____.

So carbon dioxide diffuses more _____ into the leaf from the air.

Core 4 of 5

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Tick one box in each row to show whether the statement is true or false.

Statement about diffusion	True	False
Particles move from where they are less concentrated to where they are more concentrated.		
Diffusion is faster at higher temperatures.		
A bigger difference in concentration makes diffusion faster.		
Once particles are evenly spread they stop moving.		
Leaves are thin so gases do not have to diffuse far.		

Core 5 of 5

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

4 Practical: stomata on the lower surface of a leaf.

a) Draw one stoma with its two guard cells. Label the guard cells and the pore.

Draw here. 画在这里。

b) Record the number of stomata you can see in the field of view.

Count 1	Count 2	Count 3	Mean

Challenge

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 Most leaves have far more stomata on their lower surface than on their upper surface. Suggest two reasons why this helps the plant.

C2 A student counted a mean of 32 stomata in a field of view with an area of 0.5 mm^2. The lower surface of the leaf had an area of 40 cm^2 (4000 mm^2). Estimate the number of stomata on the lower surface of the leaf. Show your working.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is the job of the guard cells?

2 Name two gases that are exchanged between a leaf and the air through its stomata.

3 Explain why a leaf loses more water on a hot day than on a cool day.

Today I can: Goal 1 ☐ Goal 2 ☐